Polynomial compression for fixed-fold sumset sizes

Draft

April 29, 2026

Abstract

For fixed h, let $N(h,k)$ be the least N such that every cardinality $|hA|$ attained by a k-element set of integers is also attained by a k-element subset of $\{1,\dots,N\}$. We prove, using Rajagopal's fixed-h theorem on the range of possible values of $|hA|$, that

$$N(h,k) \le k^{C_h}$$

for every fixed h. The new ingredient is a polynomial-diameter replacement for the sparse geometric blocks in Rajagopal's large-value construction. The replacement is carried out at the level of the low-degree Hilbert functions, which are the only data relevant to h-fold sumsets when h is fixed.

1 Introduction

For a finite set $A \subset \mathbb{Z}$ and a positive integer h, write

$$hA = \{a_1 + \cdots + a_h : a_i \in A\}$$

with repetitions allowed. Define

$$\mathcal{R}(h,k) = \{|hA| : A \subset \mathbb{Z},\ |A| = k\}.$$

Let $N(h,k)$ be the least positive integer N such that

$$\mathcal{R}(h,k) = \{|hA| : A \subseteq \{1,\dots,N\},\ |A| = k\}.$$

Equivalently, one may work in intervals $[0, N-1]$ and translate at the end.

The purpose of this note is to prove the following polynomial compression bound.

Theorem 1.1. *For every fixed $h \ge 1$ there is a constant C_h such that, for all $k \ge 2$,*

$$N(h,k) \le k^{C_h}.$$

The proof uses Rajagopal's theorem on the range of possible fixed-fold sumset sizes. Put

$$M_{h,k} = hk - h + 1, \qquad K_{h,k} = \binom{h+k-1}{h},$$

and define

$$\Delta_{h,k} = \bigcup_{\ell=0}^{\min(h,k)-3} [M_{h,k} + \ell h + 1,\ M_{h,k} + \ell h + (h-2-\ell)].$$

Here and below, $[u,v]$ denotes the set of integers n with $u \le n \le v$; if $u > v$ the interval is empty.

Theorem 1.2 (Rajagopal [1]). *For every fixed h there is k_h such that, for all $k > k_h$,*

$$\mathcal{R}(h,k) = [M_{h,k}, K_{h,k}] \setminus \Delta_{h,k}.$$

For large k, the proof is constructive and gives polynomial-diameter witnesses for all values in Rajagopal's range. The finitely many smaller values of k are absorbed at the end by increasing the constant in the polynomial bound.

The lower part of the range is already supplied by Rajagopal's filling-set construction, and its diameter is polynomial. The upper part of the range is where a change is needed. Rajagopal's large-value construction uses sparse geometric progressions; their low-degree generating functions have the right shape, but their diameters are exponential in the relevant parameters. We replace them by polynomial-diameter lattice gadgets with the same truncated generating functions. Since h is fixed, a fixed-dimensional mixed-radix map then embeds the resulting lattice sets into $\mathbb{Z}$ with only polynomial growth of the diameter.

2 Generating functions and disjoint unions

Throughout the rest of the note, h is fixed. Constants implicit in $O_h(\cdot)$, $\Theta_h(\cdot)$, and $\asymp_h$ may depend on h but not on k.

For a finite set A in a torsion-free abelian group, define the truncated sumset generating function

$$\mathcal{F}_A(z) = \sum_{q=0}^{h} |qA| z^q.$$

All identities involving $\mathcal{F}_A$ are understood modulo z^{h+1}.

Definition 2.1 (Disjoint union). *Let $A \subset \mathbb{Z}^r$ and $B \subset \mathbb{Z}^s$ be finite. Define*

$$A \sqcup B = (A,0,1,0) \cup (0,B,0,1) \subset \mathbb{Z}^{r+s+2}.$$

Iterated disjoint unions are associated arbitrarily.

Lemma 2.2. *For finite A and B,*

$$\mathcal{F}_{A\sqcup B}(z) \equiv \mathcal{F}_A(z)\mathcal{F}_B(z) \pmod{z^{h+1}}.$$

Proof. An element of $q(A\sqcup B)$ has a uniquely determined number i of summands from $(A,0,1,0)$ and $q-i$ summands from $(0,B,0,1)$, because the last two coordinates are $(i, q-i)$. Hence

$$|q(A \sqcup B)| = \sum_{i=0}^{q} |iA|\,|(q-i)B|,$$

which is exactly the coefficient identity. □

Lemma 2.3 (Mixed-radix embedding). *Let $A \subset \mathbb{Z}^D$ have all coordinates in $[0, M]$. If $Q > hM$ and*

$$\phi(x_1,\dots,x_D) = x_1 + Qx_2 + \cdots + Q^{D-1}x_D,$$

then ϕ is a Freiman isomorphism of order h on A. In particular, $|q\phi(A)| = |qA|$ for every $0 \le q \le h$, and

$$\phi(A) \subset [0, DMQ^{D-1}].$$

Thus, if $D = O_h(1)$ and $M \le k^{O_h(1)}$, then $\phi(A)$ has polynomial diameter in k.

Proof. Each coordinate of a sum of at most h elements of A lies in $[0, hM]$. Since $Q > hM$, equality of the images under ϕ is equality of the base-Q digits, and hence equality of the vector sums. $\square$

3 Polynomial-diameter free and carry gadgets

We first record an explicit polynomial-diameter supply of B_h-sets.

Lemma 3.1 (Polynomial free blocks)**.** *For every $r \geq 1$ there is a set $U_r \subset \mathbb{Z}^{h+1}$ of size r, with coordinates bounded by r^h, such that*

$$|qU_r| = \binom{r+q-1}{q} \qquad (0 \leq q \leq h).$$

Equivalently,

$$\mathcal{F}_{U_r}(z) \equiv (1-z)^{-r} \pmod{z^{h+1}}.$$

For $r = 0$ we take $U_0 = \varnothing$ and $\mathcal{F}_{U_0} = 1$.

Proof. Take

$$U_r = \{(1, i, i^2, \ldots, i^h) : 1 \leq i \leq r\}.$$

Suppose two q-fold sums are equal with $q \leq h$. The first coordinate gives that the two sums use the same number q of terms. The remaining coordinates give equality of the first h power sums of the two multisets of indices. Newton's identities then give equality of the elementary symmetric functions, and hence equality of the two multisets. Therefore all q-fold sums are distinct. $\square$

Next we construct polynomial replacements for the two sparse geometric blocks in Rajagopal's proof. We use a dissociated base.

Definition 3.2. *A sequence $u_1, \ldots, u_r$ in a torsion-free abelian group is L-dissociated if*

$$\sum_i c_i u_i = \sum_i c_i' u_i, \qquad \sum_i c_i, \sum_i c_i' \leq L,$$

with all $c_i, c_i' \in \mathbb{N}_0$, implies $c_i = c_i'$ for all i.

For our applications it is enough to take $L = h^2$. Explicitly,

$$u_i = (1, i, i^2, \ldots, i^{h^2}) \in \mathbb{Z}^{h^2+1} \qquad (1 \leq i \leq r).$$

The same Newton-identity argument as above shows that these vectors are h^2-dissociated, and their coordinates are bounded by r^{h^2}.

3.1 The G-gadget

For $3 \leq m \leq h$ and $r \geq 0$, define

$$G_{m,r} = \{0\} \cup \{u_i, mu_i : 1 \leq i \leq r\}.$$

Thus $|G_{m,r}| = 2r + 1$.

Lemma 3.3. *For $3 \le m \le h$ and $r \ge 0$,*

$$\mathcal{F}_{G_{m,r}}(z) \equiv \frac{(1-z^m)^r}{(1-z)^{2r+1}} \pmod{z^{h+1}}.$$

Moreover, defining

$$P^G_{m,r}(z) = (1-z^m)^r,$$

one has

$$P^G_{m,r}(z) - P^G_{m,r+1}(z) = z^m(1-z^m)^r.$$

Proof. Consider a q-fold sum with $q \le h$. It has the form

$$\sum_i a_i u_i + \sum_i b_i(mu_i) = \sum_i (a_i + mb_i)u_i,$$

with $a_i, b_i \ge 0$. Since $\sum_i (a_i + mb_i) \le mh \le h^2$, the coefficient vector $c_i = a_i + mb_i$ is determined by the sum.

Write $c_i = mB_i + A_i$ with $0 \le A_i < m$. The minimal number of nonzero summands needed to obtain this coefficient vector is $\sum_i (A_i + B_i)$, and copies of $0 \in G_{m,r}$ can pad this to any larger length. Therefore the generating function is

$$\frac{1}{1-z}\left(\sum_{A=0}^{m-1} z^A\right)^r \left(\sum_{B\ge 0} z^B\right)^r = \frac{(1-z^m)^r}{(1-z)^{2r+1}}.$$

The final identity is immediate. □

3.2 The H-gadget

For $2 \le m \le h-1$ and $r \ge 0$, define

$$H_{m,r} = \{u_i, mu_i : 1 \le i \le r\}.$$

Thus $|H_{m,r}| = 2r$ and $H_{m,0} = \varnothing$.

Lemma 3.4. *For $2 \le m \le h-1$ and $r \ge 0$,*

$$\mathcal{F}_{H_{m,r}}(z) \equiv \frac{P^H_{m,r}(z)}{(1-z)^{2r}} \pmod{z^{h+1}},$$

where

$$P^H_{m,r}(z) = \frac{(1-z^m)^r - z^{m-1}(1-z)^r}{1-z^{m-1}}.$$

Moreover,

$$P^H_{m,r}(z) \equiv 1 - \binom{r}{2} z^{m+1} + \sum_{\nu=m+2}^{h} O_h(r^{\nu-m+1}) z^\nu \pmod{z^{h+1}},$$

and

$$P^H_{m,r}(z) - P^H_{m,r+1}(z) \equiv rz^{m+1} + \sum_{\nu=m+2}^{h} O_h(r^{\nu-m}) z^\nu \pmod{z^{h+1}}.$$

Proof. A q-fold sum from $H_{m,r}$, with $q \le h$, has the form

$$\sum_i a_i u_i + \sum_i b_i (m u_i) = \sum_i (a_i + m b_i) u_i.$$

Again the coefficient vector $c_i = a_i + m b_i$ is determined by the sum, because $\sum_i c_i \le mh \le h^2$.

Write

$$c_i = m d_i + e_i, \qquad 0 \le e_i < m,$$

and set $D = \sum_i d_i$, $E = \sum_i e_i$. For a fixed coefficient vector c, choosing b_i with $0 \le b_i \le d_i$ gives length

$$\sum_i (a_i + b_i) = mD + E - (m-1) \sum_i b_i.$$

As the b_i vary, the sum $\sum_i b_i$ runs through every integer from 0 to D. Thus this coefficient vector contributes to exactly the degrees

$$D+E, \quad D+E+(m-1), \ldots, \quad mD+E.$$

Consequently,

$$\begin{aligned}
\mathcal{F}_{H_{m,r}}(z) &\equiv \sum_{d_i \ge 0,\ 0 \le e_i < m} z^{D+E} \left(1 + z^{m-1} + \cdots + z^{(m-1)D}\right) \\
&= \left(\frac{1-z^m}{1-z}\right)^r \frac{(1-z)^{-r} - z^{m-1}(1-z^m)^{-r}}{1-z^{m-1}} \\
&= \frac{(1-z^m)^r - z^{m-1}(1-z)^r}{(1-z)^{2r}(1-z^{m-1})}.
\end{aligned}$$

This proves the formula for $P^H_{m,r}$.

The stated expansions follow by expanding the numerator in powers of z. For the one-step difference, use

$$P^H_{m,r}(z) - P^H_{m,r+1}(z) = \frac{z^m\big((1-z^m)^r - (1-z)^r\big)}{1-z^{m-1}}.$$

The first nonzero term is rz^{m+1}, and the coefficient of z^ν is $O_h(r^{\nu-m})$ for $\nu \ge m+2$. $\square$

4 Coefficient notation

We use a truncated version of Rajagopal's coefficient notation. A series $P(z) = \sum_{q=0}^h p_q z^q$ lies in $O_h[[Xz]]$ if $p_q = O_h(X^q)$ for $0 \le q \le h$. It lies in $\Theta_{h,+}[[Xz]]$ if $p_q \asymp_h X^q$ and $p_q > 0$ for all $0 \le q \le h$.

The following elementary dominance lemma is the only analytic estimate needed below.

Lemma 4.1. *Fix $0 \le \alpha < 1$. Let $c \to \infty$, and let $E_1(z), \ldots, E_s(z)$ be a bounded number of truncated series with*

$$E_i(z) = 1 + O_h[[c^\alpha z]].$$

Then

$$(1-z)^{-c} \prod_{i=1}^s E_i(z) \in \Theta_{h,+}[[cz]].$$

More generally, if

$$D(z) = az^d(1 + O_h[[c^\alpha z]])$$

with $a > 0$ and $0 \le d \le h$, then

$$D(z)(1-z)^{-c}\prod_{i=1}^{s} E_i(z) \in az^d\Theta_{h,+}[[cz]].$$

Proof. It suffices to prove the second statement; the first is the case $a = 1$, $d = 0$. For $q \ge d$, the coefficient of z^q is

$$a\binom{c+q-d-1}{q-d} + O_h\left(a\sum_{e=1}^{q-d} c^{\alpha e}c^{q-d-e}\right).$$

The main term is $\asymp_h ac^{q-d}$, while every error term is $O_h(ac^{q-d-(1-\alpha)e}) = o_h(ac^{q-d})$. Thus the coefficient is positive and $\asymp_h ac^{q-d}$ for all sufficiently large c. $\square$

5 The lower part of the range

We quote Rajagopal's filling-set lemma in a form that records the diameter.

Definition 5.1. *A set $B \subseteq [0, d]$ is (h, d)-filling if*

$$hB = [0, hd].$$

Lemma 5.2 (Rajagopal's filling lemma [1])**.** *For fixed $h \ge 2$ and all sufficiently large k, if*

$$k - 2 \le d \le \frac{(k-1)^h}{4^{(h^2+h)/2}},$$

then there is an (h, d)-filling set $B \subseteq [0, d]$ with $|B| = k - 1$ and $[0, k/8] \subseteq B$.

Proposition 5.3 (Small values have polynomial witnesses)**.** *For fixed $h \ge 2$ there are constants $a_h > 0$ and $C_h > 0$ such that, for all sufficiently large k, every*

$$t \in [M_{h,k}, a_h k^h] \setminus \Delta_{h,k}$$

is equal to $|hA|$ for some $A \subseteq [0, C_h k^h]$ with $|A| = k$.

Proof. Take

$$D_{\max} = \left\lfloor \frac{(k-1)^h}{4^{(h^2+h)/2}} \right\rfloor.$$

Let $d \in [k-2, D_{\max}]$, let $i \in [1, h]$, and let $B \subseteq [0, d]$ be the set supplied by Lemma 5.2. For large k, the inclusion $[0, k/8] \subseteq B$ implies $[0, h^2] \subseteq B$. Define

$$A = \{0\} \cup (i + B).$$

Then $|A| = k$ and $A \subseteq [0, h + D_{\max}]$. Since $hB = [0, hd]$ and $[0, h^2] \subseteq B$, one has

$$hA = \{0\} \cup [i, h(i + d)].$$

Indeed, the interval $[i, hi]$ is contained in $i + B$, and the interval $[hi, h(i+d)]$ is $hi + hB$. Therefore

$$|hA| = h(i+d) - i + 2.$$

Write $i + d = k + \ell$. Then

$$|hA| = hk + h\ell - i + 2 = M_{h,k} + h\ell + (h - i + 1).$$

For a fixed ℓ, as i ranges over the admissible values, this gives precisely the integers in the ℓth row outside $\Delta_{h,k}$, namely

$$M_{h,k} + h\ell + [\max(1, h - \ell - 1), h].$$

As d ranges up to $D_{\max}$, this covers all values up to $a_h k^h$ outside $\Delta_{h,k}$, for some $a_h > 0$ depending only on h. The diameter is $O_h(k^h)$. □

6 The large part of the range

We now prove the polynomial-diameter replacement for Rajagopal's large-value construction.

6.1 The dense block

We need the same dense blocks as Rajagopal. For $b \geq 3$, set

$$s_b = (h-1)(b-2) + 1,$$

and for $1 \leq j \leq s_b$ define

$$B_{j,b} = [0, b-2] \cup \{h(b-2) + 2 - j\}.$$

Thus $|B_{j,b}| = b$.

Lemma 6.1 (Rajagopal's dense block [1])**.** *For fixed h and $b \geq 3$, the sets $B_{j,b}$ satisfy:*

(a) *for $2 \leq \eta \leq h$ and $2 \leq j \leq s_b$,*

$$0 \leq |\eta B_{j-1,b}| - |\eta B_{j,b}| \leq h;$$

(b) *for $1 \leq \eta \leq h$ and $1 \leq j \leq s_b$,*

$$|\eta B_{j,b}| = O_h(b);$$

(c)

$$\mathcal{F}_{B_{1,3}}(z) \equiv (1-z)^{-3} \pmod{z^{h+1}};$$

(d) *for $b > 3$,*

$$\mathcal{F}_{B_{1,b}}(z) \equiv (1-z)^{-1} \mathcal{F}_{B_{s_{b-1},b-1}}(z) \pmod{z^{h+1}}.$$

6.2 The sparse polynomial block

Fix the exponents

$$\alpha = \frac{9}{10}, \qquad \beta = \frac{4}{5}, \qquad \gamma = \frac{7}{10}.$$

They are chosen only to satisfy

$$0 < \beta < \alpha < 1, \qquad 2\beta > 1, \qquad 2\beta\gamma > 1.$$

Fix $b \in [3, k - \lfloor k^{\gamma} \rfloor]$ and put $c = k - b$. Let

$$S = \lfloor c^{\alpha} \rfloor, \qquad T = \left\lfloor c^{\beta} \right\rfloor.$$

For parameters

$$0 \le r_m \le S \quad (3 \le m \le h), \qquad 1 \le u_m \le T \quad (2 \le m \le h-1),$$

define

$$R = c - \sum_{m=3}^{h} (2r_m + 1) - \sum_{m=2}^{h-1} 2u_m.$$

For large k this is nonnegative. Define the sparse block

$$C(\mathbf{r}, \mathbf{u}) = \bigsqcup_{m=3}^{h} G_{m,r_m} \sqcup \bigsqcup_{m=2}^{h-1} H_{m,u_m} \sqcup U_R.$$

Then $|C(\mathbf{r}, \mathbf{u})| = c$, and Lemmas 2.2, 3.1, 3.3, and 3.4 give

$$\mathcal{F}_C(z) \equiv \frac{1}{(1-z)^c} \prod_{m=3}^{h} (1 - z^m)^{r_m} \prod_{m=2}^{h-1} P^H_{m,u_m}(z) \pmod{z^{h+1}}. \tag{1}$$

Lemma 6.2 (Variation of the sparse block)**.** *For fixed h and sufficiently large c, the following hold.*

(a) *If $3 \le \mu \le h$, then increasing r_μ by 1 decreases $\mathcal{F}_C(z)$ by an element of*

$$z^{\mu} \Theta_{h,+}[[cz]].$$

(b) *If $2 \le \mu \le h-1$, then increasing u_μ by 1 decreases $\mathcal{F}_C(z)$ by an element of*

$$u_\mu z^{\mu+1} \Theta_{h,+}[[cz]].$$

Proof. All products are truncated modulo z^{h+1}. The remaining factors in (1) are of the form $1 + O_h[[c^{\alpha} z]]$, since $r_m \le c^{\alpha}$ and $u_m \le c^{\beta} \le c^{\alpha}$.

For (a), by Lemma 3.3, the positive drop in the numerator is

$$z^{\mu}(1 - z^{\mu})^{r_\mu} = z^{\mu}(1 + O_h[[c^{\alpha} z]]).$$

Multiplying by the other numerator factors and by $(1-z)^{-c}$, Lemma 4.1 gives the claimed $z^{\mu}\Theta_{h,+}[[cz]]$ drop.

For (b), Lemma 3.4 gives

$$P^H_{\mu,u_\mu}(z) - P^H_{\mu,u_\mu+1}(z) = u_\mu z^{\mu+1}(1 + O_h[[c^{\alpha} z]])$$

up to degree h. Lemma 4.1 again applies. $\square$

6.3 A connectedness lemma

We use Rajagopal's interval lemma, included here for completeness.

Lemma 6.3. *Let*

$$f : [0, n_1] \times \cdots \times [0, n_d] \to \mathbb{Z}$$

be nonincreasing in each coordinate. For $1 \le i \le d$, define

$$\delta_i = \max\bigl(f(x) - f(x + e_i)\bigr),$$

where the maximum is over all x with $x_i < n_i$, and define

$$\Delta_i = \min\bigl(f(x_1, \ldots, x_{i-1}, 0, x_{i+1}, \ldots, x_d) - f(x_1, \ldots, x_{i-1}, n_i, x_{i+1}, \ldots, x_d)\bigr).$$

If $\delta_1 \le 1$ and $\delta_i \le \Delta_{i-1}$ for every $2 \le i \le d$, then the image of f is an interval of integers.

Proof. For fixed $x_2, \ldots, x_d$, varying x_1 gives an interval, because the successive drops are 0 or 1. Suppose by induction that, after varying $x_1, \ldots, x_{i-1}$ with later coordinates fixed, the image is an interval. When x_i is increased by one, the new interval starts at most δ_i below the old one, while the old interval has length at least Δ_{i-1}. Thus consecutive intervals overlap. Inducting on i proves the claim. □

6.4 Fixed b: connectedness

For fixed b and parameters as above, define

$$A = B_{j,b} \sqcup C(\mathbf{r}, \mathbf{u}), \qquad 1 \le j \le s_b.$$

Let $\mathcal{A}_b$ be the family of all such A.

Lemma 6.4. *For fixed $h \ge 3$ and all sufficiently large k, the set*

$$\{|hA| : A \in \mathcal{A}_b\}$$

is an interval of integers for every $b \in [3, k - \lfloor k^\gamma \rfloor]$.

Proof. Order the variables as

$$r_h, u_{h-1}, r_{h-1}, u_{h-2}, \ldots, r_3, u_2, j.$$

There are $2h - 3$ variables. We apply Lemma 6.3 to the function $f = |hA|$ on the corresponding box, shifting the variables so that each range starts at 0.

Because $A = B_{j,b} \sqcup C$, Lemma 2.2 gives

$$|hA| = \sum_{\eta=0}^{h} |\eta C| \, |(h - \eta) B_{j,b}|. \tag{2}$$

By Lemma 6.2 and Lemma 6.1(a), the function f is nonincreasing in every variable. Increasing r_h by one changes $\mathcal{F}_C$ only in degree h, where it decreases $|hC|$ by exactly 1; hence $\delta_1 = 1$.

It remains to compare the one-step drops δ_i with the full variations Δ_{i-1}. We record the estimates needed. If the variable is r_μ with $3 \le \mu \le h$, Lemma 6.2 and (2) give

$$\delta(r_\mu) \le C_h\big(c^{h-\mu} + bc^{h-\mu-1}\big), \tag{3}$$

where terms with negative exponents are omitted. Since the same positive leading terms occur at every step, varying r_μ over its full range gives

$$\Delta(r_\mu) \ge c_h S\big(c^{h-\mu} + bc^{h-\mu-1}\big). \tag{4}$$

Similarly, if the variable is u_μ with $2 \le \mu \le h-1$, then

$$\delta(u_\mu) \le C_h T\big(c^{h-\mu-1} + bc^{h-\mu-2}\big), \tag{5}$$

and

$$\Delta(u_\mu) \ge c_h T^2\big(c^{h-\mu-1} + bc^{h-\mu-2}\big). \tag{6}$$

Indeed, the total drop is a sum of one-step drops with weights $u_\mu = 1, 2, \dots, T-1$, whose sum is $\asymp T^2$. Finally, by Lemma 6.1(a) and (2), changing j by one changes only the dense-block factors, and

$$\delta(j) \le C_h \sum_{\eta=0}^{h-2} |\eta C| \le C_h c^{h-2}. \tag{7}$$

Now compare consecutive variables. Since $\beta < \alpha$, equations (5) and (4) imply that each u_μ-drop is at most the preceding $r_{\mu+1}$ full variation, for large c. For the reverse comparison, note that $c \ge k^\gamma + O(1)$ and $b \le k$, so the condition $2\beta\gamma > 1$ gives $b \le c^{2\beta - o(1)}$. Together with $2\beta > 1$, equations (3) and (6) imply that each r_μ-drop is at most the preceding u_μ full variation. Finally, (7) is dominated by (6) with $\mu = 2$, again because $2\beta > 1$.

Thus the hypotheses of Lemma 6.3 hold, and the image of f is an interval. □

6.5 Varying b

Proposition 6.5 (Large values have polynomial witnesses)**.** *Fix $h \ge 3$ and $\varepsilon > 0$. For all sufficiently large k, every integer*

$$t \in [\varepsilon k^h, K_{h,k}]$$

is equal to $|hA|$ for some k-element set A contained in an interval of length $k^{O_h(1)}$.

Proof. Let

$$\mathcal{A} = \bigcup_{b=3}^{k-\lfloor k^\gamma \rfloor} \mathcal{A}_b.$$

By Lemma 6.4, each $h\mathcal{A}_b := \{|hA| : A \in \mathcal{A}_b\}$ is an interval.

We first show that consecutive intervals overlap. For a fixed $b > 3$, take in $\mathcal{A}_b$ the member with all sparse parameters minimal and $j = 1$. Its sparse block has generating function $(1-z)^{-c}$. In $\mathcal{A}_{b-1}$, take all sparse parameters minimal and $j = s_{b-1}$. Its sparse block has generating function $(1-z)^{-(c+1)}$. Lemma 6.1(d) gives

$$\mathcal{F}_{B_{1,b}}(z)(1-z)^{-c} \equiv \mathcal{F}_{B_{s_{b-1},b-1}}(z)(1-z)^{-(c+1)} \pmod{z^{h+1}}.$$

Hence the two chosen sets have the same value of $|hA|$, and $h\mathcal{A}_b \cap h\mathcal{A}_{b-1} \neq \varnothing$. Therefore $h\mathcal{A}$ is an interval.

At $b = 3$, with all sparse parameters minimal and $j = 1$, Lemma 6.1(c) gives

$$\mathcal{F}_A(z) \equiv (1-z)^{-3}(1-z)^{-(k-3)} = (1-z)^{-k} \pmod{z^{h+1}}.$$

Thus

$$|hA| = \binom{h+k-1}{h} = K_{h,k}.$$

At the other end, take $b = k - \lfloor k^\gamma \rfloor$, put all sparse parameters at their maximal values, and take $j = s_b$. Then $c = \lfloor k^\gamma \rfloor$. For every member of $\mathcal{A}_b$ we have $|\eta C| = O_h(c^\eta)$ and $|qB_{j,b}| = O_h(b)$ for $q \geq 1$, while $|0B_{j,b}| = 1$. Therefore this endpoint has

$$|hA| \leq C_h(c^h + bc^{h-1}) = O_h(k^{\gamma h} + k^{1+\gamma(h-1)}) = o_h(k^h).$$

Consequently, for all sufficiently large k, the interval $h\mathcal{A}$ contains $[\varepsilon k^h, K_{h,k}]$.

It remains only to record the diameter. The dense block $B_{j,b}$ lies in an interval of length $O_h(k)$. The free block U_R has coordinates bounded by k^h. The bases used in $G_{m,r}$ and $H_{m,u}$ have coordinates bounded by $k^{O_h(1)}$, since $r, u \leq k$. There are only $O_h(1)$ disjoint-union components, so the resulting lattice set lies in fixed dimension and has coordinates bounded by $k^{O_h(1)}$. Lemma 2.3 embeds it into $\mathbb{Z}$ with diameter $k^{O_h(1)}$, preserving all q-fold sumset sizes for $q \leq h$. $\square$

7 The two-fold case

We include the elementary quadratic argument for $h = 2$, so that Theorem 1.1 is self-contained in the endpoint case not covered by the large-value construction.

Proposition 7.1. *For every $k \geq 1$,*

$$N(2,k) \leq 8k^2 + 2k + 2.$$

Proof. It is enough to work in $[0, L]$ and translate by 1 at the end. Put

$$K_n = \binom{n+1}{2}, \qquad \Delta_n = K_n - (2n-1) = \binom{n-1}{2}.$$

Every k-element set has two-fold sumset size between $2k-1$ and K_k. Conversely, fix $t \in [2k-1, K_k]$, and put $D = K_k - t$. Choose the least m such that $D \leq \Delta_m$, and set

$$s = \Delta_m - D.$$

If $m = 1$, put $B = \{0\}$. If $m \geq 2$, then $0 \leq s \leq m-2$ and we put

$$B = \{0, 1, \ldots, m-2\} \cup \{m-1+s\}.$$

Writing $L = \max B$, we have $L \leq 2m$ and

$$|B + B| = K_m - D.$$

For $m = 1$ this is immediate. For $m \geq 2$, if $P = \{0, \ldots, m-2\}$ and $x = m - 1 + s$, then $P + P = [0, 2m-4]$, $x + P = [x, x + m - 2]$, these intervals overlap or touch, and $2x$ contributes one further point, giving $|B + B| = 2m - 1 + s = K_m - D$.

Let $r = k - m$. If $r = 0$, take $A = B$ and the count is already correct. If $r = 1$, take $A = B \cup \{2L + 1\}$. Then the three pieces $B + B$, $B + \{2L + 1\}$, and $\{4L + 2\}$ are disjoint, so $|A + A| = |B + B| + m + 1 = K_{m+1} - D = t$. We may therefore assume $r \geq 2$. Choose an odd prime p with $r \leq p < 2r$, and set

$$Q = \max(2p, L + p) + 1, \qquad c_i = Qi + (i^2 \bmod p), \quad 0 \leq i < r.$$

The set $C = \{c_0, \ldots, c_{r-1}\}$ is Sidon. Indeed, $c_i + c_j = c_u + c_v$ first gives $i + j = u + v$ because $Q > 2p - 2$, and reduction modulo p then gives $i^2 + j^2 \equiv u^2 + v^2$, whence $ij \equiv uv$ and $\{i, j\} = \{u, v\}$. Also $c_{i+1} - c_i > L$, so the translates $B + c_i$ are disjoint. Put

$$M = c_{r-1} + L + 1, \qquad S = \{M + c_i : 0 \leq i < r\}, \qquad A = B \cup S.$$

Then $B + B$, $B + S$, and $S + S$ are disjoint. Moreover $|B + S| = mr$ and $|S + S| = K_r$. Hence

$$|A + A| = (K_m - D) + mr + K_r = K_k - D = t.$$

Finally, since $p < 2r$ and $L \leq 2m$,

$$\max A \leq 2r(L + 4r + 1) + L + 1 \leq 8k^2 + 2k + 1.$$

After translating by 1, the set lies in $[1, 8k^2 + 2k + 2]$. □

8 Proof of the main theorem

We now finish the proof of Theorem 1.1. The case $h = 1$ is trivial. The case $h = 2$ is Proposition 7.1.

Fix $h \geq 3$. Let a_h be as in Proposition 5.3, and apply Proposition 6.5 with $\varepsilon = a_h/2$. For all sufficiently large k, Proposition 5.3 realizes every value in

$$[M_{h,k}, a_h k^h] \setminus \Delta_{h,k}$$

inside an interval of polynomial length, while Proposition 6.5 realizes every value in

$$[(a_h/2)k^h, K_{h,k}]$$

inside an interval of polynomial length. Thus every value in

$$[M_{h,k}, K_{h,k}] \setminus \Delta_{h,k}$$

has a polynomial-diameter witness. By Rajagopal's range theorem (Theorem 1.2), this is exactly $\mathcal{R}(h, k)$ for all sufficiently large k.

For the finitely many smaller values of k, finiteness is immediate: for each attainable value choose one witnessing set, translate it into the positive integers, and take the maximum endpoint over these finitely many choices. Increasing the exponent C_h if necessary, these finitely many cases are also bounded by k^{C_h}. Finally, all constructions above were made in nonnegative intervals. Translating by $+1$ places the set in $\{1, \ldots, N\}$ and does not change $|hA|$.

This proves Theorem 1.1.

Remark 8.1 (Lower bound)**.** *The maximum value* $K_{h,k} = \binom{h+k-1}{h}$ *is attained by a* B_h*-set. If* $A \subseteq [1, N]$ *has* $|hA| = K_{h,k}$*, then* $hA \subseteq [h, hN]$*, so*

$$\binom{h+k-1}{h} \leq hN - h + 1.$$

Therefore

$$N(h, k) \gg_h k^h.$$

Thus the polynomial upper bound has the right qualitative order of growth in the exponent of k *up to the value of the exponent.*

References

[1] I. Rajagopal, *Possible Sizes of Sumsets*, arXiv:2510.23022, 2025.